Fourier Transform-2

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where

The Fourier Transform of $f(x)$ →

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(v)\, e^{-wvi}\, dv$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-wxi}\, dx.$$

v is a dummy variable, replace it with x.

The Inverse Fourier Transform of $f(x)$ →

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w)\, e^{wxi}\, dw$$

THEOREM 1 **Existence of the Fourier Transform**

If $f(x)$ is absolutely integrable on the x-axis and piecewise continuous on every finite interval, then the Fourier transform $\hat{f}(w)$ of $f(x)$ given by (6) exists.

The following conditions are sufficient for the existence of the Fourier Transform FT of a function:

i) $f(x)$ is piecewise continuous on every finite sub-interval of $\mathbb{R}$.

ii) $f(x)$ is absolutely integral on the x-axis. i.e $\int_{-\infty}^{\infty} |f(x)|\, dx$ converges.

EXAMPLE 1 **Fourier Transform**

Find the Fourier transform of $f(x) = 1$ if $|x| < 1$ and $f(x) = 0$ otherwise.

Solution. Using (6) and integrating, we obtain

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-iwx}\, dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-iwx}}{-iw}\Big|_{-1}^{1} = \frac{1}{-iw\sqrt{2\pi}} (e^{-iw} - e^{iw}).$$

As in (3) we have $e^{iw} = \cos w + i \sin w$, $e^{-iw} = \cos w - i \sin w$, and by subtraction

$$e^{iw} - e^{-iw} = 2i \sin w.$$

Substituting this in the previous formula on the right, we see that i drops out and we obtain the answer

$$\hat{f}(w) = \sqrt{\frac{\pi}{2}} \frac{\sin w}{w}.$$

■

EXAMPLE 2 **Fourier Transform**

Find the Fourier transform $\mathcal{F}(e^{-ax})$ of $f(x) = e^{-ax}$ if $x > 0$ and $f(x) = 0$ if $x < 0$; here $a > 0$.

Solution. From the definition (6) we obtain by integration

$$\mathcal{F}(e^{-ax}) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-ax} e^{-iwx}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \frac{e^{-(a+iw)x}}{-(a+iw)}\Big|_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}(a+iw)}.$$

This proves formula 5 of Table III in Sec. 11.10. ■

5) PROPERTIES OF FOURIER TRANSFORMS (FTs): 51

Let $\mathcal{F}[f] = \hat{f}(w)$ and $\mathcal{F}[g] = \hat{g}(w)$ be Fourier Transforms of f and g, respectively.

1- Linearity Property:

$$\mathcal{F}[af + bg] = a\mathcal{F}[f] + b\mathcal{F}[g], \quad a, b \in \mathbb{R}.$$

2- Shifting Property:

$$\mathcal{F}[f(x-a)] = e^{-iwa}\hat{f}(w).$$

3- Change of Scale Property:

$$\mathcal{F}[f(ax)] = \frac{1}{|a|}\hat{f}\left(\frac{w}{a}\right).$$

4- Modulation Property:

$$\mathcal{F}[f(x)\cos ax] = \frac{1}{2}[\hat{f}(w+a) + \hat{f}(w-a)]$$

Convolution

The **convolution** $f * g$ of functions f and g is defined by

(11) $$h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp = \int_{-\infty}^{\infty} f(x-p) g(p)\, dp.$$

THEOREM 4 **Convolution Theorem**

Suppose that $f(x)$ and $g(x)$ are piecewise continuous, bounded, and absolutely integrable on the x-axis. Then

(12) $$\mathcal{F}(f * g) = \sqrt{2\pi}\, \mathcal{F}(f)\, \mathcal{F}(g).$$

$$(f * g)(x) = \int_{-\infty}^{\infty} \hat{f}(w)\hat{g}(w)e^{iwx}\,dw,$$

THEOREM 3 **Fourier Transform of the Derivative of $f(x)$**

Let $f(x)$ be continuous on the x-axis and $f(x) \to 0$ as $|x| \to \infty$. Furthermore, let $f'(x)$ be absolutely integrable on the x-axis. Then

(9) $$\mathcal{F}\{f'(x)\} = iw\mathcal{F}\{f(x)\}.$$

Two successive applications of (9) give

$$\mathcal{F}(f'') = iw\mathcal{F}(f') = (iw)^2\mathcal{F}(f).$$

Since $(iw)^2 = -w^2$, we have for the transform of the second derivative of f

(10) $$\mathcal{F}\{f''(x)\} = -w^2\mathcal{F}\{f(x)\}.$$

EXAMPLE 3 **Application of the Operational Formula (9)**

Find the Fourier transform of xe^{-x^2} from Table III, Sec 11.10.

Solution. We use (9). By formula 9 in Table III

$$\begin{aligned}
\mathcal{F}(xe^{-x^2}) &= \mathcal{F}\{-\tfrac{1}{2}(e^{-x^2})'\} \\
&= -\tfrac{1}{2}\mathcal{F}\{(e^{-x^2})'\} \\
&= -\tfrac{1}{2}iw\mathcal{F}(e^{-x^2}) \\
&= -\frac{1}{2}iw\frac{1}{\sqrt{2}}e^{-w^2/4} \\
&= -\frac{iw}{2\sqrt{2}}e^{-w^2/4}.
\end{aligned}$$

Examples:

⊙ Find the Fourier Transform of $f(x) = k$ if $0 < x < a$ and $f(x) = 0$, otherwise.

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx.$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^a k e^{-i\omega x} dx$$

$$= \frac{k}{\sqrt{2\pi}} \left[\frac{e^{i\omega x}}{-i\omega} \right]_0^a$$

$$= \frac{k}{\sqrt{2\pi}} \left(\frac{1 - e^{i\omega a}}{i\omega} \right).$$

← Note that in general Fourier Transforms are complex-valued functions.

⊙ (The Gaussian) Find the Fourier Transform of e^{-ax^2}, $a > 0$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-i\omega x}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} \cdot e^{-i\omega x}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2 - i\omega x}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\sqrt{a}x + \frac{i\omega}{2\sqrt{a}}\right)^2 + \left(\frac{i\omega}{2\sqrt{a}}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}} e^{\frac{-\omega^2}{4a}} \int_{-\infty}^{\infty} e^{-\left(\sqrt{a}x + \frac{i\omega}{2\sqrt{a}}\right)^2} dx$$

Let $t = \sqrt{a}x + \frac{i\omega}{2\sqrt{a}} \Rightarrow dx = \frac{1}{\sqrt{a}}\, dt$

$$\therefore \quad \hat{f}(\omega) = \frac{e^{\frac{-\omega^2}{4a}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2} \frac{dt}{\sqrt{a}}.$$

$$= \frac{e^{\frac{-\omega^2}{4a}}}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-t^2} dt.$$

$$= \frac{e^{\frac{-\omega^2}{4a}}}{\sqrt{2\pi}} \frac{1}{\sqrt{a}} \sqrt{\pi}$$

$$= \frac{1}{\sqrt{2a}} e^{\frac{-\omega^2}{4a}}$$

Note: if $a = \frac{1}{2}$, then $f(x) = e^{-x^2/2}$ and $\hat{f}(\omega) = e^{-\omega^2/2}$. i.e.

The Fourier Transform of a Gaussian is a Gaussian.

Exercise: Show that $\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$.

2–11 FOURIER TRANSFORMS BY INTEGRATION

Find the Fourier transform of $f(x)$ (without using Table III in Sec. 11.10). Show details.

2. $f(x) = \begin{cases} e^{2ix} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

3. $f(x) = \begin{cases} 1 & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$

4. $f(x) = \begin{cases} e^{kx} & \text{if } x < 0 \quad (k > 0) \\ 0 & \text{if } x > 0 \end{cases}$

6. $f(x) = e^{-|x|} \quad (-\infty < x < \infty)$

7. $f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$

8. $f(x) = \begin{cases} xe^{-x} & \text{if } -1 < x < 0 \\ 0 & \text{otherwise} \end{cases}$

11.10 Tables of Transforms

Table I. Fourier Cosine Transforms

See (2) in Sec. 11.8.

	$f(x)$	$\hat{f}_c(w) = \mathcal{F}_c(f)$	
1	$\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}}\,\frac{\sin aw}{w}$	
2	$x^{a-1} \quad (0 < a < 1)$	$\sqrt{\frac{2}{\pi}}\,\frac{\Gamma(a)}{w^a}\cos\frac{a\pi}{2}$	($\Gamma(a)$ see App. A3.1.)
3	$e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}}\left(\frac{a}{a^2 + w^2}\right)$	
4	$e^{-x^2/2}$	$e^{-w^2/2}$	
5	$e^{-ax^2} \quad (a > 0)$	$\frac{1}{\sqrt{2a}}\,e^{-w^2/(4a)}$	
6	$x^n e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}}\,\frac{n!}{(a^2 + w^2)^{n+1}}\,\mathrm{Re}(a + iw)^{n+1}$	Re = Real part

Table II. Fourier Sine Transforms

See (5) in Sec. 11.8.

	$f(x)$	$\hat{f}_s(w) = \mathcal{F}_s(f)$	
1	$\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}}\left[\frac{1 - \cos aw}{w}\right]$	
2	$1/\sqrt{x}$	$1/\sqrt{w}$	
3	$1/x^{3/2}$	$2\sqrt{w}$	
4	$x^{a-1} \quad (0 < a < 1)$	$\sqrt{\frac{2}{\pi}}\,\frac{\Gamma(a)}{w^a}\sin\frac{a\pi}{2}$	($\Gamma(a)$ see App. A3.1.)
5	$e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}}\left(\frac{w}{a^2 + w^2}\right)$	
6	$\frac{e^{-ax}}{x} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}}\arctan\frac{w}{a}$	

Table III. Fourier Transforms

See (6) in Sec. 11.9.

	$f(x)$	$\hat{f}(w) = \mathcal{F}(f)$
1	$\begin{cases} 1 & \text{if } -b < x < b \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}}\,\frac{\sin bw}{w}$
2	$\begin{cases} 1 & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$	$\frac{e^{-ibw} - e^{-icw}}{iw\sqrt{2\pi}}$
3	$\frac{1}{x^2 + a^2} \quad (a > 0)$	$\sqrt{\frac{\pi}{2}}\,\frac{e^{-a\lvert w\rvert}}{a}$
4	$\begin{cases} x & \text{if } 0 < x < b \\ 2x - b & \text{if } b < x < 2b \\ 0 & \text{otherwise} \end{cases}$	$\frac{-1 + 2e^{ibw} - e^{-2ibw}}{\sqrt{2\pi}w^2}$
5	$\begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (a > 0)$	$\frac{1}{\sqrt{2\pi}(a + iw)}$
6	$\begin{cases} e^{ax} & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$	$\frac{e^{(a-iw)c} - e^{(a-iw)b}}{\sqrt{2\pi}(a - iw)}$